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Motivational Speech Synthesis

by Luis Küffner, Pierre-Louis Suckrow, Patrick Stecher and Nikolaj Wolff

Introduction

Amid the increasing popularity of fitness and entrepreneurship in Western subcultures, clips of *motivational speech* receive millions of views across social media. Usually, these audiovisual artifacts show male business leaders, authors, and other influential figures narrating strategies for success. Paired with epic, emotional background music, these videos function as vehicles for self-motivation and goal pursuit. With a primarily male target group, success is often tied to wealth, professional growth, or appeal to women, while obstructed by weakness, fragility, or discontinuity. Motivational speech thus emerges from a society of self-optimization and an ethos of constant productivity, self-isolation, competition, and meritocracy.

To illustrate the work ethic reflected in motivational speech, examples of motivational quotes identified across various social media platforms include the following:

- "do or die"
- "do it alone"
- "don't complain, enjoy your pain"
- "nobody is coming to save you"
- "become dangerous"
- "reject weakness, embrace discipline"
- "nobody cares, work harder"
- "take it personal"
- "kill the boy"

Focusing on the human voice as the primary medium within this audiovisual subculture, its characteristic prosodic patterns play a decisive role in the appearance and perception of motivational speech. With *Motivational Speech Synthesis*, we therefore aim to

1. replicate those prosodic features by introducing a novel approach to control them through a one-dimensional scale
2. while creating a space for artistic reflection to extract the underlying attitudes of this subculture as a whole.

Since both motivational speech and machine learning techniques build on generalization and an anticipated forward process, their correlation allows us to engage both through mutual artistic reflection. Motivated by this and inspired by advances in emotional speech synthesis, we developed a novel framework for emotional controlled text-to-speech (TTS) generation built on web-scraped motivational speech.

Emotional control is achieved through the one-dimensional *motivational factor*, enabling fine-grained intensity control over prosody during inference. Through dimensionality reduction, we map a three-dimensional emotional representation of speech onto a one-dimensional scale from 0 (low *motivational factor*) to 1 (high *motivational factor*).

Reflecting the notion of social mobility found within capitalist ideologies such as motivational speech subcultures, the proposed *motivational factor* aligns with attitudes like "The harder you work, the more you can get". Despite this asserted emphasis on individual determination, which is also manifested by the American Dream, OECD (Organisation for Economic Co-operation and Development) data suggests that income, education, and occupational status are still rather strongly shaped by one's family background[1]. Given this, a child born into a low-income family in the US would take an estimated five generations to earn average income[2]. Rather than exposing people consuming motivational speech, we reveal and criticize the circumstances and mindsets behind narratives that arrive as symptoms of a society driven by self-optimization, growth, and success. By questioning how we define "success", *Motivational Speech Synthesis* addresses our work ethic and how we approach goals and challenges in life.

Related Work

Although many state-of-the-art models in emotion-aware text-to-speech and speech emotion recognition—such as XTTS-v2[3], MetaVoice[4], Parler-TTS [5], or StyleTTS 2 [6]—produce high-quality speech, few generate speech with specific emotional inflections. Voice cloning has reached a high level of sophistication, yet integrating prosodic variation into TTS systems remains critical for synthesized, human-like speech.

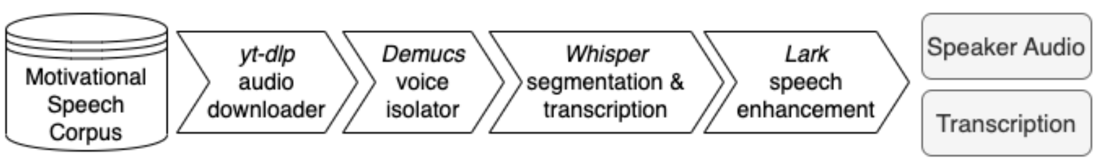
The model proposed by Cho et al.[7] allows emotion intensity control along with style transfer, while EmoKnob[8] provides fine-grained emotion modulation by extending the voice cloning-based TTS model *MetaVoice*. The authors constructed a speaker representation space for systematic modeling and comparison of speaker characteristics. An emotional embedding is created by extracting an emotion direction vector, obtained by computing the normalized difference between the emotional and neutral embeddings of the same speaker. Subsequently, this embedding is added to the speaker representation space.

One accessible model that explores transformer-based architectures for improving SER by embedding analyzed speech into a 3D emotional space is a fine-tuned version of wav2vec 2.0[9]. Another approach we examined is emotion2vec [10], which provides a high-dimensional speech emotion representation model in addition to a SER foundation model classifying emotions into discrete categories. Due to limited availability of labeled data for emotion recognition[11], both models use self-supervised learning frameworks. Here, a common approach uses pretrained self-supervised models, such as wav2vec 2.0[12], trained on large-scale speech datasets and fine-tuned for emotion recognition tasks[13]. This methodology overcomes data scarcity by utilizing rich representations learned from vast amounts of unlabeled speech data, thereby improving SER performance.

Method

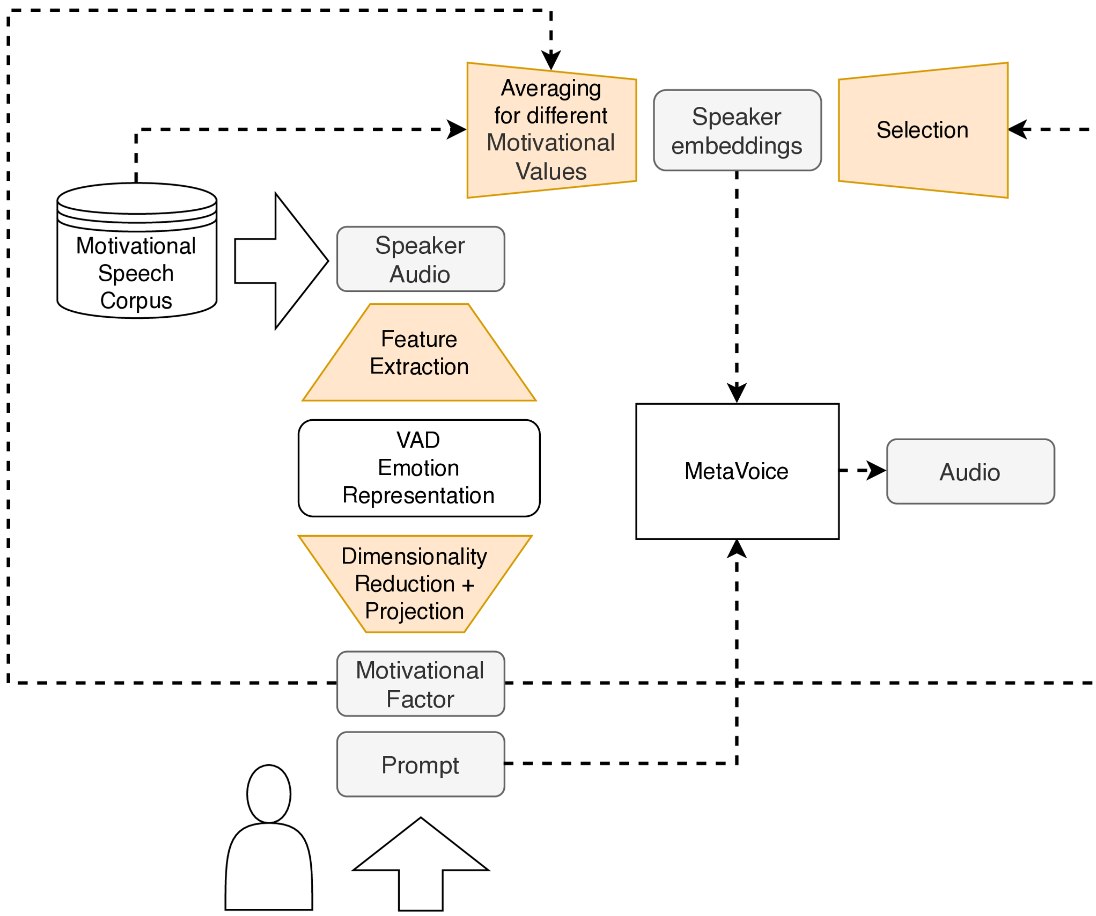
Preprocessing

Since artistically reflecting motivational speech through its prosody is specific in both technical domain and data sourcing, we needed a dataset accurately representing motivational speech as a subcultural artifact. Given this, the absence of professional training data, and our non-commercial artistic interests, we sourced data from social media platforms like YouTube, where motivational speeches exhibit a consistent structure, often including dramatic instrumental music. To capture only the speech, we employed a multi-stage preprocessing pipeline (see Figure 1). After collecting YouTube audio dedicated to motivational content, speech components were isolated using the music source separation algorithm Demucs[14]. Once separated, the extracted speech undergoes further refinement, including speech enhancement via ai-coustics' proprietary model called *Lark*[15] and transcription via Whisper[16]. During development, we compiled a motivational speech dataset comprising 414,024 data points with a total duration of approximately 371 hours.



Overview of the data processing pipeline.

Model architecture



Proposed model architecture with EmoKnob TTS model that uses our motivational factor directly at inference and a reference audio averaged from our motivational speech corpus

After evaluating text-to-speech (TTS) models capable of emotional conditioning, we based our architecture on established approaches. Many contemporary models use higher-dimensional emotional representations, such as those produced by the aforementioned SER models[17], to generate expressive speech. This allows implementation of our *motivational factor* without developing a new TTS architecture. With appropriate dimensionality reduction, higher-dimensional emotional representations can be mapped onto a one-dimensional *motivational factor*. Our factor ranges continuously from 0, indicating low motivational prosody intensity, to 1, indicating high motivational prosody intensity. The derived *motivational factor* can then serve indirectly as a user-specified input condition during inference.

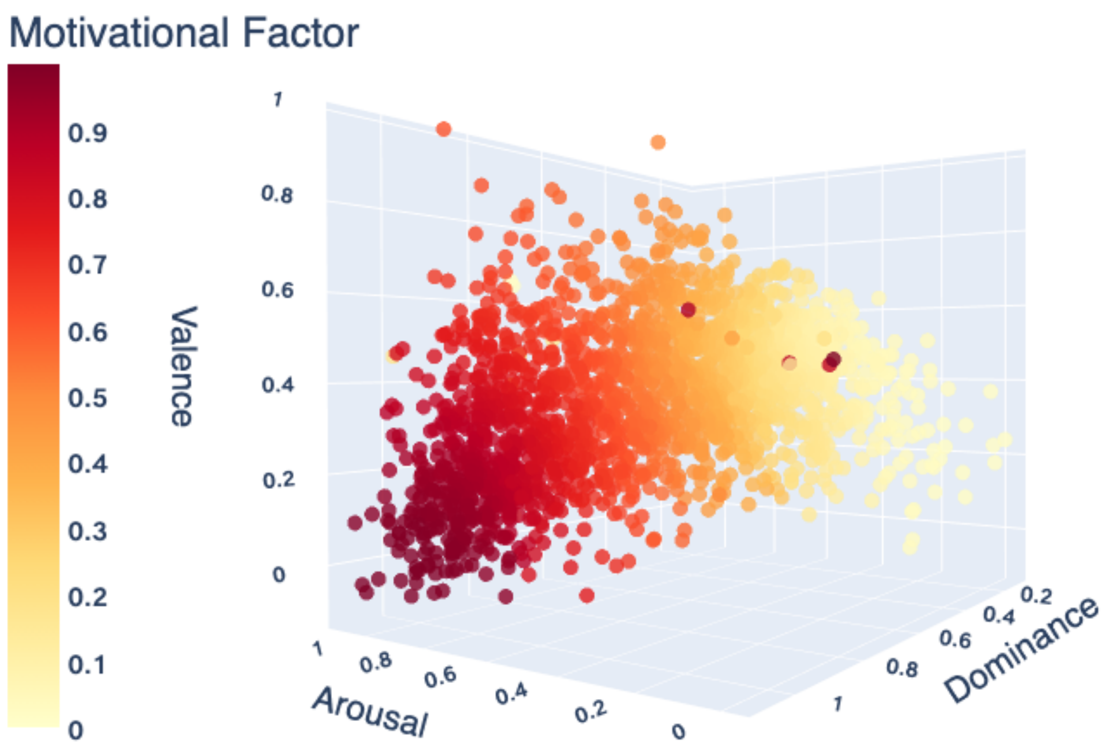
In the domain of SER, emotions are primarily represented in two ways: as discrete categories (e.g., happy, sad, angry) [18] or as positions in a continuous emotion space usually defined by three dimensions: valence, arousal, and dominance. The scales within this 3D emotion model range from negative to positive emotions (valence), calm to stimulated emotions (arousal), and submissive to dominant emotions (dominance)[19]. As our proposed *motivational factor* does not fit into any of these discrete categories, but rather lives within the continuous space, our research focused on architectures that operate with this approach.

With the three-dimensional *VAD* space as our high-dimensional representation, we projected our motivational speech corpus onto it using the inference model proposed by Wagner et al.[20]. To represent our *motivational factor* as a single dimension, we therefore reduced these three dimensions into one by applying the UMAP (Uniform Manifold Approximation and Projection)[21] algorithm, resulting in the desired projection ranging from 0 to 1.

In our approach, speaker embeddings, which encode speaker style within a high-dimensional feature space, are provided during inference to guide synthesis. By mapping distinct speaker embeddings to their fitting *motivational factor* and averaging them to discrete steps, these embeddings can also indirectly represent different *motivational factors*. During inference, an averaged speaker embedding nearest to the specified motivational input value is selected to guide speech generation (see Figure 2). In our implementation, we adopted EmoKnob[22] as our TTS model and computed averaged speaker embeddings in increments of 0.05. For each increment, a representative speaker embedding was obtained by calculating the mean of the *k*-nearest neighbor (*k*NN$=400$) embeddings within the speaker embedding space.

Discussion and Conclusion

Our research introduced and evaluated a method for synthesizing motivational speech using averaged speaker embeddings within a modified existing model architecture. By compressing dimensional emotional relationships into a single motivational intensity scale, the method enables adjustment of motivational prosody in synthesized speech. Our EmoKnob-based architecture generates averaged speaker embeddings, constituting a lightweight modification of the existing MetaVoice model without computationally intensive training or fine-tuning. We hope that our proposed architecture will inspire future research in emotion-specific speech synthesis across various tasks and domains.



Visualization of 2000 dataset audio points embedded into VAD space. Motivational factor representation via colormap.

Figure 3 visualizes a subset of n=2000 audio data points that were randomly selected along the *motivational factor* dimension and projected into VAD space. The distribution follows a curvilinear trajectory from lower valence, arousal, and dominance toward higher arousal and dominance. This latent structure corresponds to the pattern captured by UMAP and reduced to a single dimension, supporting its use for representing the data along one *motivational factor*.

Now what to do with these technical achievements? Nothing is won when motivational speech and its disruptive attitudes towards work and success are merely reproduced through a novel TTS model. Artistic tension emerges when disclosure and attitude collide. Through exposure and exaggeration of the absurdity of motivational speech, the work acts as a critical mirror to societies and subcultures fueling self-exploitation, self-isolation, and toxic forms of masculinity for so-called work-life success, which appears rather illusory. By parameterizing emotional persuasion, motivation appears not as an authentic force but as a reproducible sociocultural symbol. This deconstruction is achieved through a demonstration video, which conveys the visual and auditive language of motivational speech videos, while the narrated content reveals a fact-based counter-perspective. Through the appropriation of the subculture from its broadcast channels and its reinsertion into the same cultural space through short video content across social media, we examine its transformation of social and economic pressure into personal affect. Imperatives such as “nobody cares, work harder”, “reject weakness, embrace discipline”, and “don’t complain, enjoy your pain” falsely suggest a direct correlation of social mobility to relentless commitment, effort, and quantity of work. While critical factors for social mobility are not individual hard work paired with continuity and dominance but rather family background, class, and access, alternative and less extreme approaches to working culture and goal pursuit prevent the long-term erosion of bodily and psychological well-being. In return, weakness, fragility, and discontinuity are not flaws but part of being human, encouraging conscious work rather than constant productivity.

A project webpage[23] accompanies this work, providing the artistic demonstration video, audio examples, an interactive visualization, a ready-to-run Collab demo, and access to the code repository[24] containing the model source code.

We gratefully acknowledge Berlin University of the Arts for supplying the computational resources, as well as ai-coustics for providing their speech-enhancement model to improve our dataset.

Luis Küffner is a sound and media artist, abstracting underlying patterns and sociocultural contexts of technologies, collective dynamics, and symbolic systems. Rearticulating compositional strategies from electroacoustic music into dense, hypnotic, temporal forms and physical objects, he investigates the digital voice in generative and algorithmic synthesis processes. His installations, musical performances, and electronic pieces have been presented across venues and festivals, such as Kunstpalast Düsseldorf, silent green Berlin, International Computer Music Conference Seoul, Goethe-Insitute Bangkok, and Blurred Edges Festival Hamburg.

Pierre-Louis Suckrow is an artist, researcher, and software developer. In his work, he recomposes cultural artefacts through code, moving image, sound, and other time-based media. In doing so, he explores, extracts, and interprets their immanent structures and phenomena in order to develop methods that can be transferred between technical, artistic, and scientific practice.

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